UNIT-II

TOP DOWN PARSING:

- Top-down parsing can be viewed as the problem of constructing a parse tree for the given input string, starting from the root and creating the nodes of the parse tree in preorder (depth-first left to right).
- Equivalently, top-down parsing can be viewed as finding a leftmost derivation for an input string.

It is classified in to two different variants namely; one which uses Back Tracking and the other is Non Back Tracking in nature.

Non Back Tracking Parsing: There are two variants of this parser as given below.

1. **Table Driven Predictive Parsing :**
   i. LL (1) Parsing
2. **Recursive Descent parsing**

Back Tracking

1. **Brute Force method**

NON BACK TRACKING:

LL (1) Parsing or Predictive Parsing

LL (1) stands for, left to right scan of input, uses a Left most derivation, and the parser takes 1 symbol as the look ahead symbol from the input in taking parsing action decision.

A non recursive predictive parser can be built by maintaining a stack explicitly, rather than implicitly via recursive calls. The parser mimics a leftmost derivation. If w is the input that has been matched so far, then the stack holds a sequence of grammar symbols a such that

$$S \overset{*}{\underset{lm}{\Rightarrow}} w\alpha$$

The table-driven parser in the figure has

- An input buffer that contains the string to be parsed followed by a $ Symbol, used to indicate end of input.
- A stack, containing a sequence of grammar symbols with a $ at the bottom of the stack, which initially contains the start symbol of the grammar on top of $.
- A parsing table containing the production rules to be applied. This is a two dimensional array M [Non terminal, Terminal].
- A parsing Algorithm that takes input String and determines if it is conformant to Grammar and it uses the parsing table and stack to take such decision.

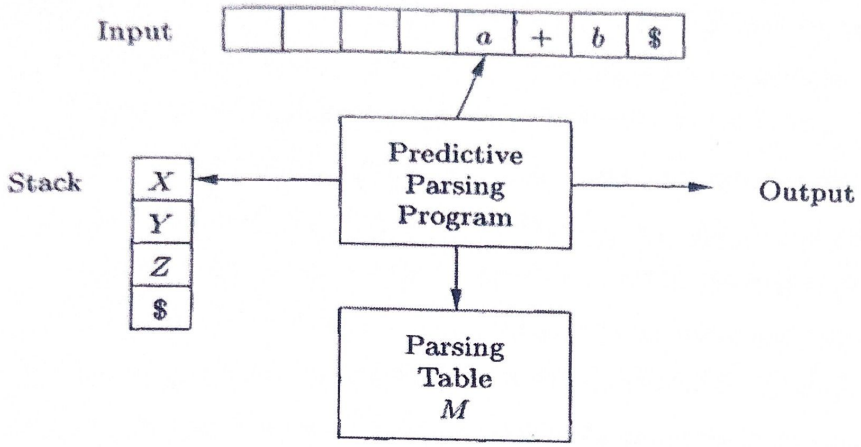


Figure 2.2: Model for table driven parsing

The Steps Involved In constructing an LL(1) Parser are:

1. Write the Context Free grammar for given input String
2. Check for Ambiguity. If ambiguous remove ambiguity from the grammar
3. Check for Left Recursion. Remove left recursion if it exists.
4. Check For Left Factoring. Perform left factoring if it contains common prefixes in more than one alternates.
5. Compute FIRST and FOLLOW sets
6. Construct LL(1) Table
7. Using LL(1) Algorithm generate Parse tree as the Output

Context Free Grammar (CFG): CFG used to describe or denote the syntax of the programming language constructs. The CFG is denoted as G, and defined using a four tuple notation.

Let G be CFG, then G is written as, G= (V, T, P, S)

Where

- Σ V is a finite set of Non terminal; Non terminals are syntactic variables that denote sets of strings. The sets of strings denoted by non terminals help define the language generated by the grammar. Non terminals impose a hierarchical structure on the language that is key to syntax analysis and translation.

- Σ T is a Finite set of Terminal; Terminals are the basic symbols from which strings are formed. The term "token name" is a synonym for "terminal" and frequently we will use the word "token" for terminal when it is clear that we are talking about just the token name. We assume that the terminals are the first components of the tokens output by the lexical analyzer.

- Σ S is the Starting Symbol of the grammar, one non terminal is distinguished as the start symbol, and the set of strings it denotes is the language generated by the grammar. P is finite set of Productions; the productions of a grammar specify the manner in which the

terminals and non terminals can be combined to form strings, each production is in α->β form, where α is a single non terminal, β is (VUT)*.Each production consists of:

(a) A non terminal called the head or left side of the production; this production defines some of the strings denoted by the head.

(b) The symbol ->. Some times: = has been used in place of the arrow.

(c) A body or right side consisting of zero or more terminals and non-terminals. The components of the body describe one way in which strings of the non terminal at the head can be constructed.

Σ Conventionally, the productions for the start symbol are listed first.

Example: Context Free Grammar to accept Arithmetic expressions.

The terminals are +, *, -, (,), **id**.

The **Non terminal symbols** are **expression**, **term**, **factor** and expression is the starting symbol.

expression	⟶	***expression + term***
expression	⟶	***expression – term***
expression	⟶	***term***
term	⟶	***term * factor***
term	⟶	***term / factor***
term	⟶	***factor***
factor	⟶	***(expression)***
factor	⟶	***id***

Figure 2.3 : Grammar for Simple Arithmetic Expressions

Notational Conventions Used In Writing CFGs:

To avoid always having to state that "these are the terminals," "these are the non terminals," and so on, the following notational conventions for grammars will be used throughout our discussions.

1. These symbols are terminals:

(a) Lowercase letters early in the alphabet, such as a, b, e.

(b) Operator symbols such as +, *, and so on.

(c) Punctuation symbols such as parentheses, comma, and so on.

(d) The digits 0, 1. . . 9.

(e) Boldface strings such as id or if, each of which represents a single terminal symbol.

2. These symbols are non terminals:

(a) Uppercase letters early in the alphabet, such as A, B, C.

(b) The letter S, which, when it appears, is usually the start symbol.

(c) Lowercase, italic names such as *expr* or *stmt*.

(d) When discussing programming constructs, uppercase letters may be used to represent Non terminals for the constructs. For example, non terminal for expressions, terms, and factors are often represented by E, T, and F, respectively.

Using these conventions the grammar for the arithmetic expressions can be written as

E → E + T | E – T | T
T → T * F | T / F | F
F → (E) | id

DERIVATIONS:

The construction of a parse tree can be made precise by taking a derivational view, in which productions are treated as rewriting rules. Beginning with the start symbol, each rewriting step replaces a Non terminal by the body of one of its productions. This derivational view corresponds to the top-down construction of a parse tree as well as the bottom construction of the parse tree.

Σ Derivations are classified in to **Let most Derivation** and **Right Most Derivations.**

Left Most Derivation (LMD):

It is the process of constructing the parse tree or accepting the given input string, in which at every time we need to rewrite the production rule it is done with left most non terminal only.

Ex: - If the Grammar is **E-> E+E | E*E | -E| (E) | id** and the input string is **id + id* id**

The production **E -> - E** signifies that if E denotes an expression, then – E must also denote an expression. The replacement of a single E by - E will be described by writing

E => -E which is read as **"E derives _E"**

For a general definition of derivation, consider a non terminal A in the middle of a sequence of grammar symbols, as in αAβ, where α and β are arbitrary strings of grammar symbol. Suppose A ->γ is a production. Then, we write αAβ => αγβ. The symbol => means "derives in one step". Often, we wish to say, "Derives in zero or more steps." For this purpose, we can use the symbol $\overset{*}{\Rightarrow}$, If we wish to say, "Derives in $\overset{+}{\Rightarrow}$ one or more steps." We cn use the symbol $\overset{+}{\Rightarrow}$. If S $\overset{*}{\Rightarrow}$ a, where S is the start symbol of a grammar G, we say that α is a sentential form of G.

The Leftmost Derivation for the given input string **id + id* id** is

E => E +E

=> id + **E**

=> id + **E** * E

=> id + id * **E**

=> id + id * id

NOTE: Every time we need to start from the root production only, the under line using at Non terminal indicating that, it is the non terminal (left most one) we are choosing to rewrite the productions to accept the string.

Right Most Derivation (RMD):

It is the process of constructing the parse tree or accepting the given input string, every time we need to rewrite the production rule with Right most Non terminal only.

The Right most derivation for the given input string **id + id* id** is

E => E + E

=> E + E * E

=> E + E * id

=> E + id * id

=> id + id * id

NOTE: Every time we need to start from the root production only, the under line using at Non terminal indicating that, it is the non terminal (Right most one) we are choosing to rewrite the productions to accept the string.

What is a Parse Tree?

A parse tree is a graphical representation of a derivation that filters out the order in which productions are applied to replace non terminals.

- Σ Each interior node of a parse tree represents the application of a production.
- Σ All the interior nodes are Non terminals and all the leaf nodes terminals.
- Σ All the leaf nodes reading from the left to right will be the output of the parse tree.
- Σ If a node n is labeled X and has children n1,n2,n3,…nk with labels X1,X2,…Xk respectively, then there must be a production A->X1X2…Xk in the grammar.

Example1:- Parse tree for the input string - **(id + id) using** the above Context free Grammar is

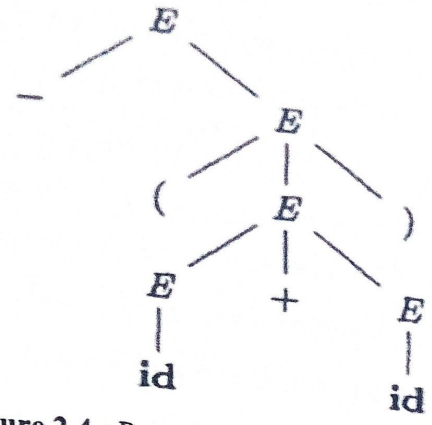


Figure 2.4 : Parse Tree for the input string - **(id + id)**

The Following figure shows step by step construction of parse tree using CFG for the parse tree for the input string - **(id + id).**

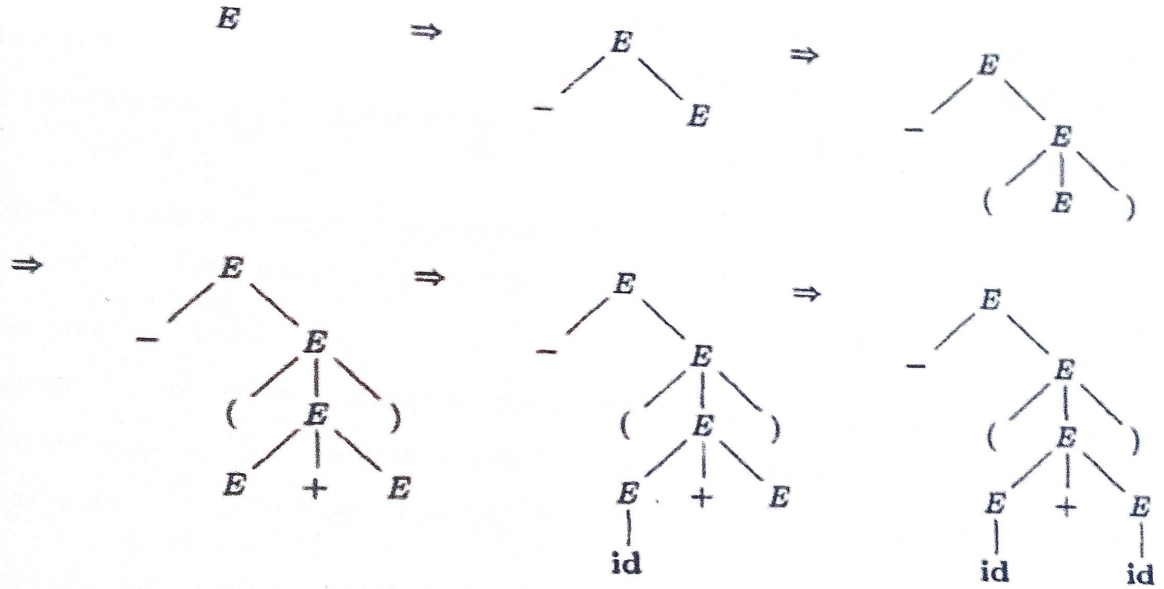


Figure 2.5 : Sequence outputs of the Parse Tree construction process for the input string **–(id+id)**

Example2:- Parse tree for the input string **id+id*id** using the above Context free Grammar is

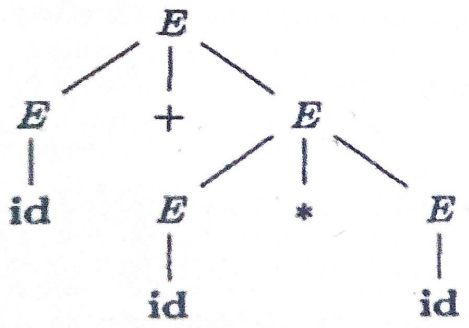


Figure 2.6: Parse tree for the input string id+ id*id

AMBIGUITY in CFGs:

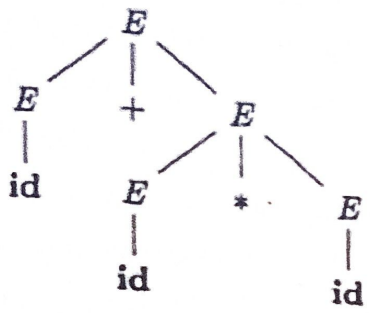
Definition: A grammar that produces more than one parse tree for some sentence (input string) is said to be ambiguous.

In other words, an ambiguous grammar is one that produces more than one leftmost derivation or more than one rightmost derivation for the same sentence.

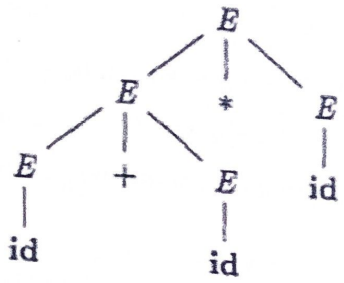
Or If the right hand production of the grammar is having two non terminals which are exactly same as left hand side production Non terminal then it is said to an ambiguous grammar.

Example : If the Grammar is **E-> E+E | E*E | -E| (E) | id** and the Input String is **id + id* id**

Two parse trees for given input string are



(a)



(b)

Two Left most Derivations for given input String are :

E => E +E
=> id + E
=> id + E * E
=> id + id * E
=> id + id * id

(a)

E => E * E
=> E+E*E
=> id + E * E
=> id+ id* E
=> id + id * id

(b)

The above Grammar is giving two parse trees or two derivations for the given input string so, it is an ambiguous Grammar

Note: LL (1) parser will not accept the ambiguous grammars or We cannot construct an LL(1) parser for the ambiguous grammars. Because such grammars may cause the Top Down parser to go into infinite loop or make it consume more time for parsing. If necessary we must remove all types of ambiguity from it and then construct.

ELIMINATING AMBIGUITY: Since Ambiguous grammars may cause the top down Parser go into infinite loop, consume more time during parsing.

Therefore, sometimes an ambiguous grammar can be rewritten to eliminate the ambiguity. The general form of ambiguous productions that cause ambiguity in grammars is

$$A \rightarrow A\alpha \mid \beta$$

This can be written as (introduce one new non terminal in the place of second non terminal)

$$A \rightarrow \beta A'$$
$$A' \rightarrow \alpha A' \mid \varepsilon$$

Example : Let the grammar is **E⟶ E+E | E*E | -E| (E) | id** . It is shown that it is ambiguous that can be written as

E → E+E
E → E-E
E → E*E
E → -E
E → (E)
E → id

In the above grammar the 1st and 2nd productions are having ambiguity. So, they can be written as

E-> E+E | E*E this production again can be written as
E-> E+E | β , where β is E*E
The above production is same as the general form. so, that can be written as
E->E+T|T
T->β

The value of β is E*E so, above grammar can be written as

1) **E->E+T|T**
2) **T-> E*E** **The first production is free from ambiguity** and substitute E->T in the 2nd production then it can be written as
   T-> T*T | -E| (E) | **id** this production again can be written as
   T->T*T | β where β is -E| (E) | id, introduce new non terminal in the Right hand side production then it becomes
   T->T*F | F
   F-> -E | (E) | id now the entire grammar turned in to it equivalent unambiguous,

The Unambiguous grammar equivalent to the given ambiguous one is

1) **E → E + T | T**
2) **T → T * F | F**
3) **F → -E | (E) | id**

LEFT RECURSION:

Another feature of the CFGs which is not desirable to be used in top down parsers is left recursion. A grammar is left recursive if it has a non terminal A such that there is a derivation A=>Aα for some string α in (TUV)*. LL(1) or Top Down Parsers can not handle the Left Recursive grammars, so we need to remove the left recursion from the grammars before being used in Top Down Parsing.

The General form of Left Recursion is

$$A \rightarrow A\alpha \mid \beta$$

The above left recursive production can be written as the non left recursive equivalent :

$$A \rightarrow \beta A'$$
$$A' \rightarrow \alpha A' \mid \epsilon$$

Example : - Is the following grammar left recursive? If so, find a non left recursive grammar equivalent to it.

$$E \rightarrow E + T \mid T$$
$$T \rightarrow T * F \mid F$$
$$F \rightarrow -E \mid (E) \mid id$$

Yes ,the grammar is left recursive due to the first two productions which are satisfying the general form of Left recursion, so they can be rewritten after removing left recursion from $E \rightarrow E + T$, and $T \rightarrow T * F$ is

$$E \rightarrow TE'$$
$$E' \rightarrow +T\,E' \mid \epsilon$$
$$T \rightarrow F\,T'$$
$$T' \rightarrow *F\,T' \mid \epsilon$$
$$F \rightarrow (E) \mid id$$

LEFT FACTORING:

Left factoring is a grammar transformation that is useful for producing a grammar suitable for predictive or top-down parsing. A grammar in which more than one production has common prefix is to be rewritten by factoring out the prefixes.

For example, in the following grammar there are n A productions have the common prefix α, which should be removed or factored out without changing the language defined for A.

$$A \rightarrow \alpha A1 \mid \alpha A2 \mid \alpha A3 \mid \alpha A4 \mid \ldots \mid \alpha An$$

We can factor out the α from all n productions by adding a new A production $A \rightarrow \alpha A'$, and rewriting the A′ productions grammar as

$$A \rightarrow \alpha A'$$
$$A' \rightarrow A1 \mid A2 \mid A3 \mid A4 \ldots \mid An$$

FIRST and FOLLOW:

The construction of both top-down and bottom-up parsers is aided by two functions, FIRST and FOLLOW, associated with a grammar G. During top down parsing, FIRST and FOLLOW allow us to choose which production to apply, based on the next input (look a head) symbol.

Computation of FIRST:

FIRST function computes the set of terminal symbols with which the right hand side of the productions begin. To compute FIRST (A) **for all grammar symbols,** apply the following rules until no more terminals or € can be added to any FIRST set.

1. If A is a terminal, then FIRST {A} = {A}.
2. If A is a Non terminal and A->X1X2…Xi
   FIRST(A)=FIRST(X1) if X1is not null, if X1 is a non terminal and X1->€, add FIRST(X2) to FIRST(A), if X2-> € add FIRST(X3) to FIRST(A), … if Xi->€ , i.e., all Xi's for i=1..i are null, add € FIRST(A).
3. If A ->€ is a production, then add € to FIRST (A).

Computation Of FOLLOW:

Follow (A) is nothing but the set of terminal symbols of the grammar that are immediately following the Non terminal A. If **a** is to the immediate right of non terminal A, then Follow(A)= {**a**}. To compute FOLLOW (A) for **all non terminals** A, apply the following rules until no more symbols can be added to any FOLLOW set.

1. Place $ in FOLLOW(S), where *S* is the start symbol, and $ is the input right end marker.
2. If there is a production A-> αBβ, then everything in FIRST (β) except € is in FOLLOW(B).
3. If there is a production A->αB or a production A-> αBβ with FIRST(β) contains €, then FOLLOW (B) = FOLLOW (A).

Example: - Compute the FIRST and FOLLOW values of the expression grammar

1. **E → TE′**
2. **E′→+TE′ | €**
3. **T → FT′**
4. **T′ →*FT′ | €**
5. **F → (E) | id**

Computing FIRST Values:

FIRST (E) = FIRST (T) = FIRST (F) = {(, id}
FIRST (E′) = {+, €}
FIRST (T′) = {*, €}

follow(E) = ~~First(T)~~ = {$,) }
follow(E′) = follow(E)
= {$,) }
follow(T) = First(E′) ∪ follow(E′))
= (+, follow(E′))
(+, $,)
follow(T′) = follow(T)
(+ $,)
follow (F) = {first(T′) ∪ follow(T)
{*, + $)}

Computing FOLLOW Values:

FOLLOW (E) = { $,), } Because it is the start symbol of the grammar.

FOLLOW (E′) = {FOLLOW (E)} satisfying the 3rd rule of FOLLOW()
= { $,) }

FOLLOW (T) = { FIRST E′} It is Satisfying the 2nd rule.
U { FOLLOW(E′) }
= {+, FOLLOW (E′)}
= { +, $,) }

FOLLOW (T′) = { FOLLOW(T)} Satisfying the 3rd Rule
= { +, $,) }

FOLLOW (F) = { FIRST (T′) } It is Satisfying the 2nd rule.
U { FOLLOW(E′) }
= {*, FOLLOW (T)}
= { *, +, $,)}.

NON TERMINAL	FIRST	FOLLOW
E	{ (, id }	{ $,) }
E′	{ +, € }	{ $,) }
T	{ (, id}	{ +, $,) }
T′	{*, €}	{ +, $,) }
F	{ (, id}	{ *, +, $,) }

Table 2.1: FIRST and FOLLOW values

Constructing Predictive Or LL (1) Parse Table:

It is the process of placing the all productions of the grammar in the parse table based on the FIRST and FOLLOW values of the Productions.

The rules to be followed to Construct the Parsing Table (M) are :

1. For Each production A-> α of the grammar, do the bellow steps.
2. For each terminal symbol 'a' in FIRST (α), add the production A-> α to M [A, a].
3. i. If € is in FIRST (α) add production A->α to M [A, b], where b is all terminals in FOLLOW (A).
   ii. If € is in FIRST(α) and $ is in FOLLOW (A) then add production A->α to M [A, $].
4. Mark other entries in the parsing table as error .

NON-TERMINALS	INPUT SYMBOLS					
	+	*	(	)	id	$

E			E TE′		E id	
E′	E′ +TE′			E′ €		E′ €
T			T FT′		T FT′	
T′	T′ €	T′ *FT′		T′ €		T′ €
F			F (E)		F id	

Table 2.2: LL (1) Parsing Table for the Expressions Grammar

Note: if there are no multiple entries in the table for single a terminal then grammar is accepted by LL(1) Parser.

LL (1) Parsing Algorithm:

The parser acts on basis on the basis of two symbols

i. A, the symbol on the top of the stack
ii. a, the current input symbol

There are three conditions for A and 'a', that are used fro the parsing program.

1. If A=a=$ then parsing is Successful.
2. If A=a≠$ then parser pops off the stack and advances the current input pointer to the next.
3. If A is a Non terminal the parser consults the entry M [A, a] in the parsing table. If M[A, a] is a Production A-> $X_1X_2..X_n$, then the program replaces the A on the top of the Stack by $X_1X_2..X_n$ in such a way that X_1 comes on the top.

STRING ACCEPTANCE BY PARSER:

If the input string for the parser is **id + id * id,** the below table shows how the parser accept the string with the help of Stack.

Stack	Input	Action	Comments
$E	id+ id* id $	E TE`	E on top of the stack is replaced by TE`
$E`T	id+ id*id $	T FT`	T on top of the stack is replaced by FT`
$E`T`F	id+ id*id $	F id	F on top of the stack is replaced by id
$E`T`id	id+ id*id $	pop and remove id	Condition 2 is satisfied
$E`T`	+id*id$	T` €	T` on top of the stack is replaced by €
$E`	+id*id$	E` +TE`	E` on top of the stack is replaced by +TE`
$E`T+	+id*id$	Pop and remove +	Condition 2 is satisfied
$E`T	id*id$	T FT`	T on top of the stack is replaced by FT`
$E`T`F	id*id$	F id	F on top of the stack is replaced by id
$E`T`id	id * id$	pop and remove id	Condition 2 is satisfied

$E`T`	*id$	T` *FT`	T` on top of the stack is replaced by *FT`
$E`T`F*	*id$	pop and remove *	Condition 2 is satisfied
$E`T`F	id$	F id	F on top of the stack is replaced by id
$E`T`id	id$	Pop and remove id	Condition 2 is satisfied
$E`T`	$	T` €	T` on top of the stack is replaced by €
$E`	$	E` €	E` on top of the stack is replaced by €
$	$	Parsing is successful	Condition 1 satisfied

Table2.3 : Sequence of steps taken by parser in parsing the input **token stream id+ id* id**

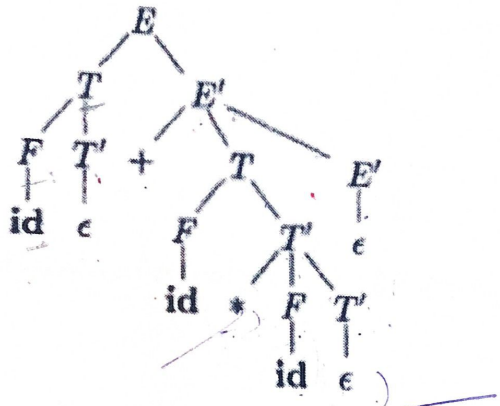


Figure 2.7: Parse tree for the input id + id* id

ERROR HANDLING (RECOVERY) IN PREDICTIVE PARSING:

In table driven predictive parsing, it is clear as to which terminal and Non terminals the parser expects from the rest of input. An error can be detected in the following situations:

1. When the terminal on top of the stack does not match the current input symbol.
2. when Non terminal A is on top of the stack, a is the current input symbol, and M[A, a] is empty or error

The parser recovers from the error and continues its process. The following error recovery schemes are use in predictive parsing:

Panic mode Error Recovery :

It is based on the idea that when an error is detected, the parser will skips the remaining input until a synchronizing token is en countered in the input. Some examples are listed below:

1. For a Non Terminal A, place all symbols in FOLLOW (A) are adde into the synchronizing set of non terminal A. For Example, consider the assignment statement "c=;" Here, the expression on the right hand side is missing. So the Follow of this is considered. It is ";" and is taken as synchronizing token. On encountering it, parser emits an error message "Missing Expression".
2. For a Non Terminal A, place all symbols in FIRST (A) are adde into the synchronizing set of non terminal A. For Example, consider the assignment statement "22c= a + b;" Here, FIRST (expr) is 22. It is ";" and is taken as synchronizing token and then the reports the error as "extraneous token".

Phrase Level Recovery :

It can be implemented in the predictive parsing by filling up the blank entries in the predictive parsing table with pointers to error Handling routines. These routines can insert, modify or delete symbols in the input.

RECURSIVE DESCENT PARSING :

A recursive-descent parsing program consists of a set of recursive procedures, one for each non terminal. Each procedure is responsible for parsing the constructs defined by its non terminal, Execution begins with the procedure for the start symbol, which halts and announces success if its procedure body scans the entire input string.

If the given grammar is

E → TE′
E′ → +TE′ | €
T → FT′
T′ → *FT′ | €
F → (E) | id

Reccursive procedures for the recursive descent parser for the given grammar are given below.

```
procedure E( )
{
        T( );
        E′( );
}
procedure T ( )
{
        F( );
        T′( );
}
Procedure E′( )
{
        if input = '+'
        {
                advance( );
                T ( );
                E′( );
                return true;
        }
        else error;
}
procedure T′( )
{
        if input = '*'
        {
                advance( );
                F ( );
```

```
                T'( );
        return true;
        }
        else return error;
}
procedure F( )
{
        if input = '('
        {
                advance( );
                E ( );
                if input = ')'
                advance( );
                return true;
        }
        else if input = "id"
        {

                advance( );
                return true;
        }
        else return error;
}
advance()
{
        input = next token;
}
```

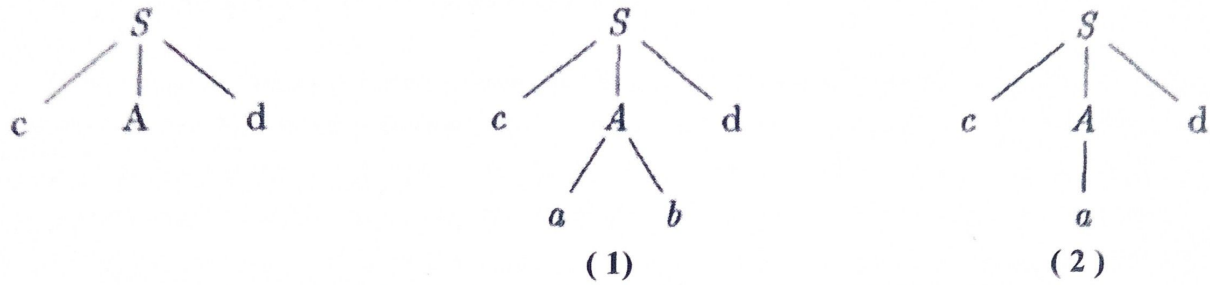
BACK TRACKING: This parsing method uses the technique called Brute Force method during the parse tree construction process. This allows the process to go back (back track) and redo the steps by undoing the work done so far in the point of processing.

Brute force method: It is a Top down Parsing technique, occurs when there is more than one alternative in the productions to be tried while parsing the input string. It selects alternatives in the order they appear and when it realizes that something gone wrong it tries with next alternative.

For example, consider the grammar bellow.

S → cAd
A → ab | a

To generate the input string "cad", initially the first parse tree given below is generated. As the string generated is not "cad", input pointer is back tracked to position "A", to examine the next alternate of "A". Now a match to the input string occurs as shown in the 2nd parse trees given below.



(1) (2)

IMPORTANT AND EXPECTED QUESTIONS

1. Explain the components of working of a Predictive Parser with an example?
2. What do the FIRST and FOLLOW values represent? Give the algorithm for computing FIRST n FOLLOW of grammar symbols with an example?
3. Construct the LL (1) Parsing table for the following grammar?
   E → E+T|T
   T → T*F
   F → (E) | id
4. For the above grammar construct, and explain the Recursive Descent Parser?
5. What happens if multiple entries occurring in your LL (1) Parsing table? Justify your answer? How does the Parser

ASSIGNMENT QUESTIONS

1. Eliminate the Left recursion from the below grammar?
   A-> Aab | AcB|b
   B-> Ba | d
2. Explain the procedure to remove the ambiguity from the given grammar with your own example?
3. Write the grammar for the if-else statement in the C programming and check for the left factoring?
4. Will the Predictive parser accept the ambiguous Grammar justify your answer?
5. Is the grammar G = { S->L=R, S->R, R->L, L->*R | **id** } an LL(1) grammar?

BOTTOM-UP PARSING

Bottom-up parsing corresponds to the construction of a parse tree for an input string beginning at the leaves (the bottom nodes) and working up towards the root (the top node). It involves "reducing an input string 'w' to the Start Symbol of the grammar. in each reduction step, a perticular substring matching the right side of the production is replaced by symbol on the left of that production and it is the Right most derivation. For example consider the following Grammar:

E → E+T|T

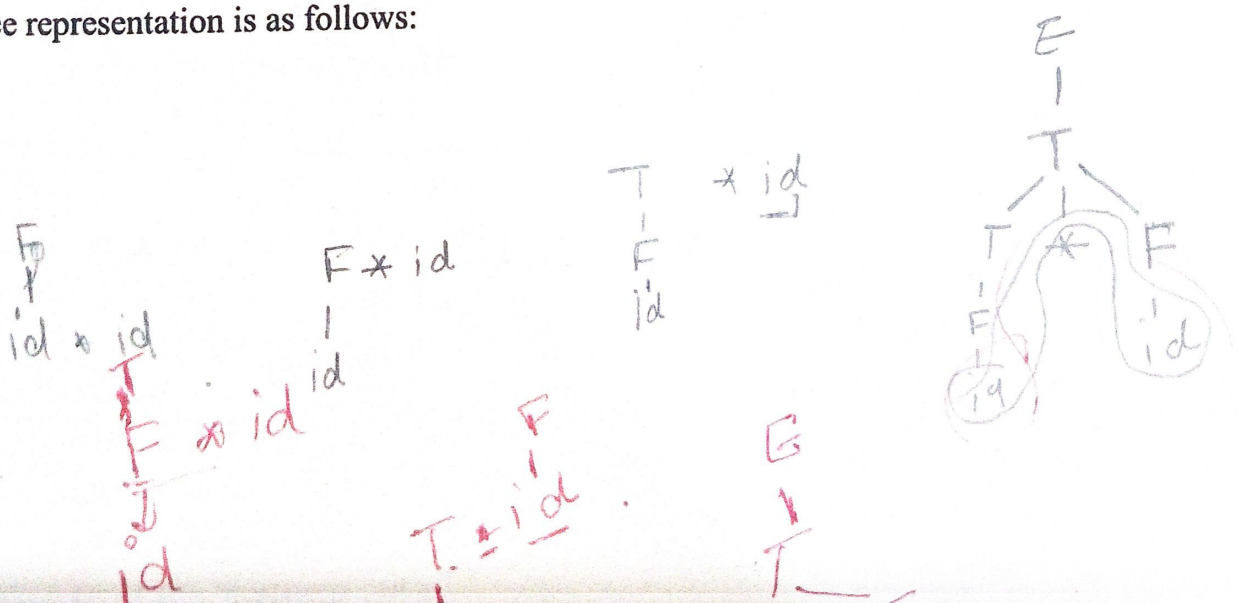
T → T*F / F

F → (E)|id

Bottom up parsing of the input string **"id * id "**is as follows:

INPUT STRING	SUB STRING	REDUCING PRODUCTION
id*id	Id	F->id
F*id	T	F->T
T*id	Id	F->id
T*F	*	T->T*F
T	T*F	E->T
E		Start symbol. Hence, the input String is accepted

Parse Tree representation is as follows:



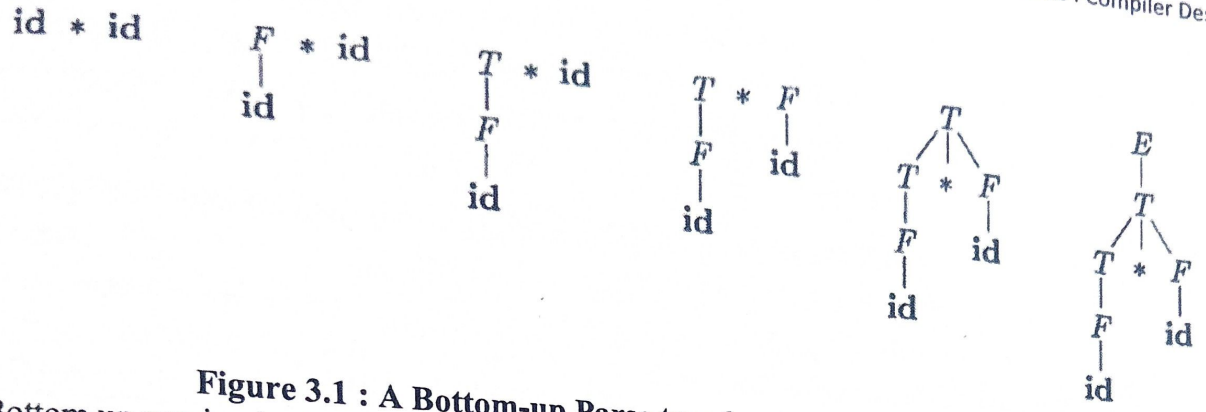


Figure 3.1 : A Bottom-up Parse tree for the input String "id*id"

Bottom up parsing is classified in to 1. Shift-Reduce Parsing, 2. Operator Precedence parsing , and 3. [Table Driven] L R Parsing

i. SLR(1)
ii. CALR (1)
iii.LALR(1)

SHIFT-REDUCE PARSING:

Shift-reduce parsing is a form of bottom-up parsing in which a stack holds grammar symbols and an input buffer holds the rest of the string to be parsed, We use $ to mark the bottom of the stack and also the right end of the input. And it makes use of the process of shift and reduce actions to accept the input string. Here, the parse tree is Constructed bottom up from the leaf nodes towards the root node.

When we are parsing the given input string, if the match occurs the parser takes the reduce action otherwise it will go for shift action. And it can accept ambiguous grammars also.

For example, consider the below grammar to accept the input string "id * id", using S-R parser

E → E+T|T
T → T*F | F
F → (E)|id

Actions of the Shift-reduce parser using Stack implementation

STACK	INPUT	ACTION
$	Id*id$	Shift
$id	*id$	Reduce with F→id
$F	*id$	Reduce with T→F
$T	*id$	Shift
$T*	id$	Shift
$T*id	$	Reduce with F→id
$T*F	$	Reduce with T→T*F
$T	$	Reduce with E→T
$E	$	Accept

Consider the following grammar:

S → aAcBe
A → Ab|b
B → d

Let the input string is "abbcde". The series of shift and reductions to the start symbol are as follows.

abbcde ⇨ aAbcde ⇨ aAcde ⇨ aAcBe ⇨ S

Note: in the above example there are two actions possible in the second Step, these are as follows :

1. Shift action going to 3rd Step
2. Reduce action, that is A->b

If the parser is taking the 1st action then it can successfully accepts the given input string, if it is going for second action then it can't accept given input string. This is called shift reduce conflict. Where, S-R parser is not able take proper decision, so it not recommended for parsing.

OPERATOR PRECEDENCE PARSING:

Operator precedence grammar is kinds of shift reduce parsing method that can be applied to a small class of operator grammars. And it can process ambiguous grammars also.

- An operator grammar has two important characteristics:
  1. There are no ϵ productions.
  2. No production would have two adjacent non terminals.
- The operator grammar to accept expressions is give below:

  E → E+E / E → E-E / E → E*E / E → E/E / E → E^E / E → -E / E → (E) / E → id

Two main Challenges in the operator precedence parsing are:

1. Identification of Correct handles in the reduction step, such that the given input should be reduced to starting symbol of the grammar.
2. Identification of which production to use for reducing in the reduction steps, such that we should correctly reduce the given input to the starting symbol of the grammar.

Operator precedence parser consists of:

1. An input buffer that contains string to be parsed followed by a$, a symbol used to indicate the ending of input.
2. A stack containing a sequence of grammar symbols with a $ at the bottom of the stack.
3. An operator precedence relation table O, containing the precedence ralations between the pair of terminal. There are three kinds of precedence relations will exist between the pair of terminal pair 'a' and 'b' as follows:
4. The relation a<•b implies that the terminal 'a' has lower precedence than terminal 'b'.
5. The relation a•>b implies that the terminal 'a' has higher precedence than terminal 'b'.
6. The relation a=•b implies that the terminal 'a' has lower precedence than terminal 'b'.

7. An operator precedence parsing program takes an input string and determines whether it conforms to the grammar specifications. It uses an operator precedence parse table and stack to arrive at the decision.

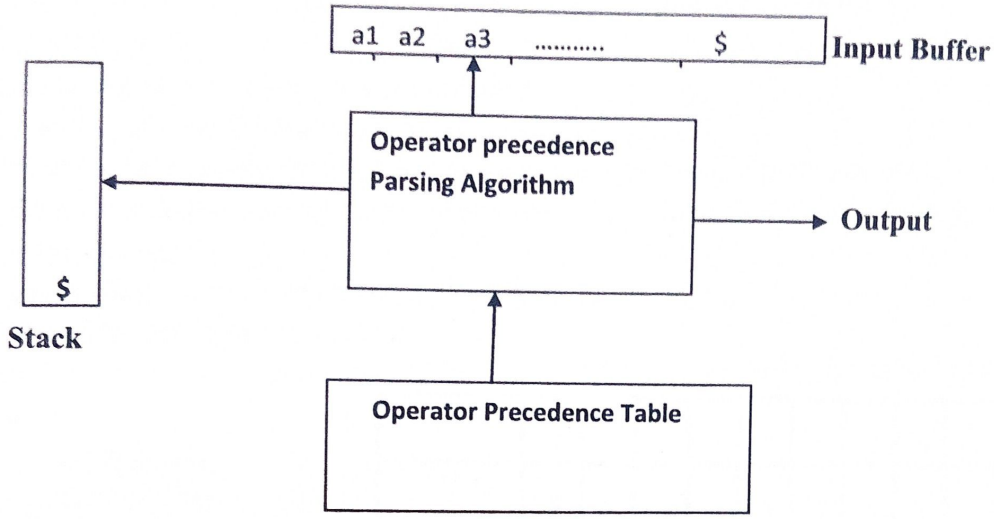


Figure3.2: Components of operator precedence parser

Example, If the grammar is

E → E+E
E → E-E
E → E*E
E → E/E
E → E^E
E → -E
E → (E)
E → id , Construct operator precedence table and accept input string **" id+id*id"**

The precedence relations between the operators are
(id) > (^) > (* /) > (+ -) > $, '^' operator is Right Associative and reaming all operators are Left Associative

	+	-	*	/	^	id	(	)	$
+	•>	•>	<•	<•	<•	<•	<•	•>	•>
-	•>	•>	<•	<•	<•	<•	<•	•>	•>
*	•>	•>	•>	•>	<•	<•	<•	•>	•>
/	•>	•>	•>	•>	<•	<•	<•	•>	•>
^	•>	•>	•>	•>	<•	<•	<•	•>	•>
Id	•>	•>	•>	•>	•>	Err	Err	•>	•>
(	<•	<•	<•	<•	<•	<•	<•	=	Err
)	•>	•>	•>	•>	•>	Err	Err	•>	•>
$	<•	<•	<•	<•	<•	<•	<•	Err	Err

The intention of the precedence relations is to delimit the handle of the given input String with <• marking the left end of the Handle and •> marking the right end of the handle.

Parsing Action:

To locate the handle following steps are followed:

1. Add $ symbol at the both ends of the given input string.
2. Scan the input string from left to right until the right most •> is encountered.
3. Scan towards left over all the equal precedence's until the first <• precedence is encountered.
4. Every thing between <• and •> is a handle.
5. $ on S means parsing is success.

Example, Explain the parsing Actions of the OPParser for the input string is **"id*id"** and the grammar is:

E → E+E
E → E*E
E → id

1. $ <• id •> *<• id•> $

The first handle is 'id' and match for the 'id 'in the grammar is E → id . So, id is replaced with the Non terminal E. the given input string can be written as

2. $ <• E •> *<• id•> $

The parser will not consider the Non terminal as an input. So, they are not considered in the input string. So , the string becomes

3. $ <• *<• id•> $

The next handle is 'id' and match for the 'id 'in the grammar is E → id . So, id is replaced with the Non terminal E. the given input string can be written as

4. $ <• *<• E•> $

The parser will not consider the Non terminal as an input. So, they are not considered in the input string. So, the string becomes

5. $ <• * •> $

The next handle is '*' and match for the ' 'in the grammar is E → E * E . So, id is replaced with the Non terminal E. the given input string can be written as

6. $ E $

The parser will not consider the Non terminal as an input. So, they are not considered in the input string. So, the string becomes

7. $ $

$ On $ means parsing successful.

Operator Parsing Algorithm:

The operator precedence Parser parsing program determines the action of the parser depending on

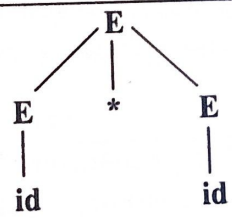
1. 'a' is top most symbol on the Stack
2. 'b' is the current input symbol

There are 3 conditions for 'a' and 'b' that are important for the parsing program

1. a=b=$, the parsing is successful
2. a <• b or a = b, the parser shifts the input symbol on to the stack and advances the input pointer to the next input symbol.
3. a •> b, parser performs the reduce action. The parser pops out elements one by one from the stack until we find the current top of the stack element has lower precedence than the most recently popped out terminal.

Example, the sequence of actions taken by the parser using the stack for the input string "id * id " and corresponding Parse Tree are as under.

STACK	INPUT	OPERATIONS
$	id * id $	$ <• id, shift 'id' in to stack
$ id	*id $	id •> *, reduce 'id' using E-> id
$E	*id $	$ <• *, shift '*' in to stack
$E*	id$	* <• id , shift 'id' in to Stack
$E*id	$	id •> $, reduce 'id' using E->id
$E*E	$	*•> $, reduce '*' using E->E*E
$E	$	$=$=$, so parsing is successful



Advantages and Disadvantages of Operator Precedence Parsing:

The following are the advantages of operator precedence parsing

1. It is simple and easy to implement parsing technique.
2. The operator precedence parser can be constructed by hand after understanding the grammar. It is simple to debug.

The following are the disadvantages of operator precedence parsing:

1. It is difficult to handle the operator like '-'which can be either unary or binary and hence different precedence's and associativities.
2. It can parse only a small class of grammar.

3. New addition or deletion of the rules requires the parser to be re written.
4. Too many error entries in the parsing tables.

LR Parsing:

Most prevalent type of bottom up parsing is LR (k) parsing. Where, L is left to right scan of the given input string, R is Right Most derivation in reverse and K is no of input symbols as the Look ahead.

- ∑ It is the most general non back tracking shift reduce parsing method
- ∑ The class of grammars that can be parsed using the LR methods is a proper superset of the class of grammars that can be parsed with predictive parsers.
- ∑ An LR parser can detect a syntactic error as soon as it is possible to do so, on a left to right scan of the input.

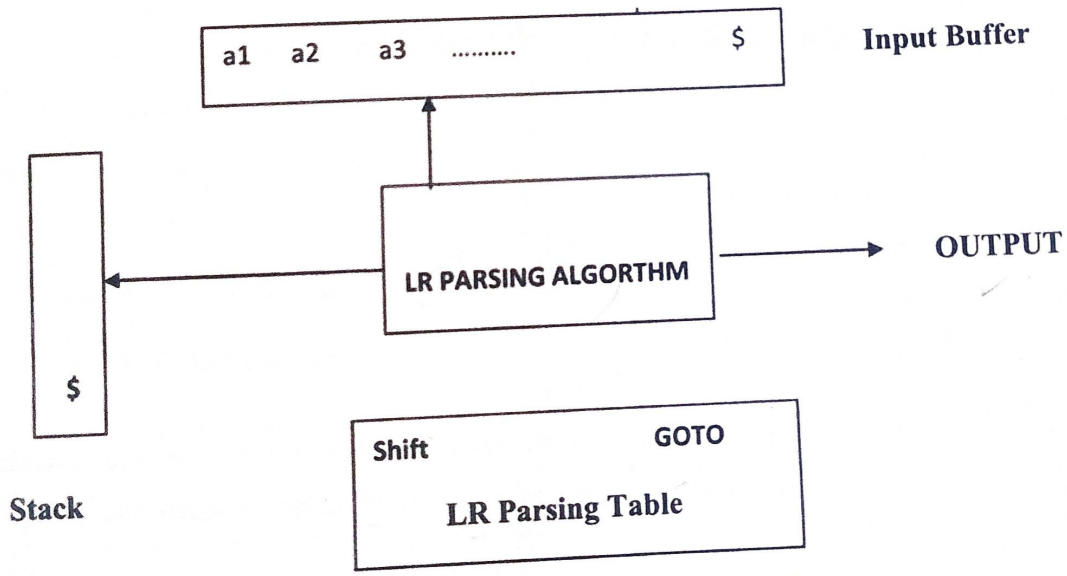


Figure 3.3: Components of LR Parsing

LR Parser Consists of

- ∑ An input buffer that contains the string to be parsed followed by a $ Symbol, used to indicate end of input.
- ∑ A stack containing a sequence of grammar symbols with a $ at the bottom of the stack, which initially contains the Initial state of the parsing table on top of $.
- ∑ A parsing table (M), it is a two dimensional array M[state, terminal or Non terminal] and it contains two parts

1. **ACTION Part**
   The ACTION part of the table is a two dimensional array indexed by state and the input symbol, i.e. **ACTION**[state][input], An action table entry can have one of following four kinds of values in it. They are:
   1. Shift X, where X is a State number.
   2. Reduce X, where X is a Production number.
   3. Accept, signifying the completion of a successful parse.
   4. Error entry.
2. **GO TO Part**
   The GO TO part of the table is a two dimensional array indexed by state and a Non terminal, i.e. GOTO[state][NonTerminal]. A GO TO entry has a state number in the table.

Σ A parsing Algorithm uses the current State X, the next input symbol 'a' to consult the entry at action[X][a]. it makes one of the four following actions as given below:

1. If the action[X][a]=shift Y, the parser executes a shift of Y on to the top of the stack and advances the input pointer.
2. If the action[X][a]= reduce Y (Y is the production number reduced in the State X), if the production is Y->β, then the parser pops 2*β symbols from the stack and push Y on to the Stack.
3. If the action[X][a]= accept, then the parsing is successful and the input string is accepted.
4. If the action[X][a]= error, then the parser has discovered an error and calls the error routine.

The parsing is classified in to

1. LR (0)
2. Simple LR (1)
3. Canonical LR (1)
4. Look ahead LR (1)

LR (1) Parsing: Various steps involved in the LR (1) Parsing:

1. Write the Context free Grammar for the given input string
2. Check for the Ambiguity
3. Add Augment production
4. Create Canonical collection of LR (0) items
5. Draw DFA
6. Construct the LR (0) Parsing table
7. Based on the information from the Table, with help of Stack and Parsing algorithm generate the output.

Augment Grammar

2. If a production X→α•Aβ is in I, then add Production which are starting with X in the State I. Rule 2 is applied until no more productions added to the State I(meaning that the • is followed by a Terminal symbol).

Example :

0. E`→ E
1. E→ E+T
2. T→ F
3. T→ T*F
4. F→ (E)
5. F→ id

LR (0) items for the Grammar is

E`→• E
E →•E+T
T→ •F
T→ •T*F
F→ • (E)
F→ • id

Closure (I_0) State

Add E `→ • E in I_0 State

Since, the '•' symbol in the Right hand side production is followed by A Non terminal E. So, add productions starting with E in to Io state. So, the state becomes

E` •E →

0. E→ •E+T
1. T→ •F

The 1st and 2nd productions are satisfies the 2nd rule. So, add productions which are starting with E and T in I_0

Note: once productions are added in the state the same production should not added for the 2nd time in the same state. So, the state becomes

0. E`→ •E
1. E→ •E+T
2. T→ •F
3. T→ •T*F
4. F→ •(E)
5. F→ • id

GO TO Operation

Go to (I_0, X), where I_0 is set of items and X is the grammar Symbol on which we are moving the '•' symbol. It is like finding the next state of the NFA for a give State I_0 and the input symbol is X. For example, if the production is E →E+T

Go to (I_0, E) is E`→ •E, E→ E•+T

Note: Once we complete the Go to operation, we need to compute closure operation for the output production

I_i

I_i

For Example, Construct the LR (0) parsing Table for the given Grammar (G)

$S \rightarrow aB$

$B \rightarrow bB \mid b$

Sol: 1. Add Augment Production and insert '•' symbol at the first position for every production in G

0. $S' \rightarrow \bullet S$
1. $S \rightarrow \bullet aB$
2. $B \rightarrow \bullet bB$
3. $B \rightarrow \bullet b$

I_0 State:

1. Add Augment production to the I_0 State and Compute the Closure

I_0 = Closure ($S' \rightarrow \bullet S$)

Since '•' is followed by the Non terminal, add all productions starting with S in to I_0 State. So, the I_0 State becomes

$I_0 =$ $S' \rightarrow \bullet S$

$S \rightarrow \bullet aB$ Here, in the S production '.' Symbol is followed by a terminal value so close the state.

I_1= Go to (I_0, S)

$S` \rightarrow S\bullet$

Closure($S` \rightarrow S\bullet$**)** = $S' \rightarrow S\bullet$ Here, The Production is reduced so close the State.

$I_1 = S' \rightarrow S\bullet$

I_2= Go to (I_0, a) = closure ($S \rightarrow a\bullet B$)

Here, the '•' symbol is followed by The Non terminal B. So, add the productions which are Starting B.

$I_2 =$ $B \rightarrow \bullet bB$

$B \rightarrow \bullet b$ Here, the '•' symbol in the B production is followed by the terminal value. So, Close the State.

$I_2 =$ $S \rightarrow a\bullet B$

$B \rightarrow \bullet bB$

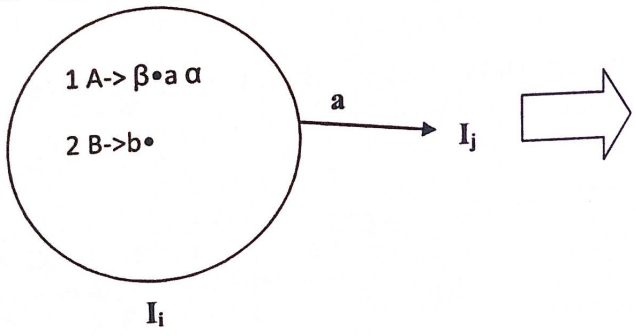
LR Parsing Table:

States	ACTION			GOTO	
	a	b	$	S	B
I_0	S_2			1	
I_1			ACC		
I_2		S_4			3
I_3	R_1	R_1	R_1		
I_4	R_3	S_4/R_3	R_3		5
I_5	R_2	R_2	R_2		

Note: if there are multiple entries in the LR (1) parsing table, then it will not accepted by the LR(1) parser. In the above table I_3 row is giving two entries for the single terminal value 'b' and it is called as Shift- Reduce conflict.

Shift-Reduce Conflict in LR (0) Parsing: Shift Reduce Conflict in the LR (0) parsing occurs when a state has

1. A Reduced item of the form A → α• and
2. An incomplete item of the form A → β•aα as shown below:

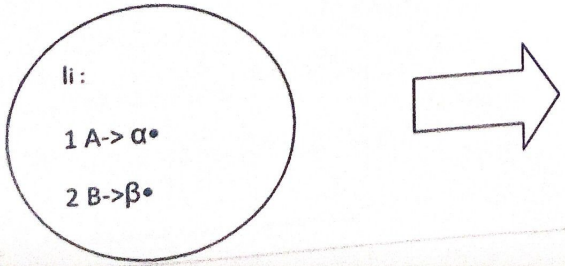


States	Action		GOTO	
	a	$	A	B
I_i	Sj/r2	r_2		
I_j				

Reduce - Reduce Conflict in LR (0) Parsing:

Reduce- Reduce Conflict in the LR (1) parsing occurs when a state has two or more reduced items of the form

1. **A → α•**
2. **B → β•** as shown below:



States	Action		GOTO	
	a	$	A	B
I_i	$r_1/r2$	r_1/r_2		

SLR PARSER CONSTRUCTION: What is SLR (1) Parsing

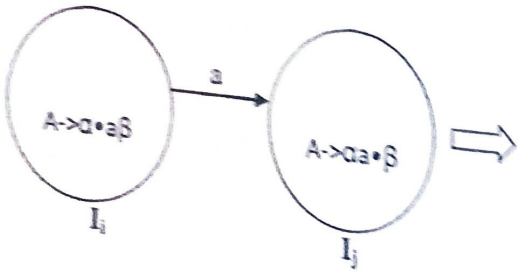
Various steps involved in the SLR (1) Parsing are:

1. Write the Context free Grammar for the given input string
2. Check for the Ambiguity
3. Add Augment production
4. Create Canonical collection of LR (0) items
5. Draw DFA
6. Construct the SLR (1) Parsing table
7. Based on the information from the Table, with help of Stack and Parsing algorithm generate the output.

SLR (1) Parsing Table Construction

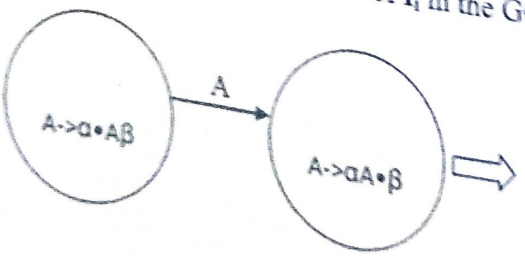
Once we have Created the canonical collection of LR (0) items, need to follow the steps mentioned below:

If there is a transaction from one state (I_i) to another state(I_j) on a terminal value then we should write the shift entry in the action part as shown below:



States	ACTION		GO TO
	a	$	A
I_i	S_j		
I_j			

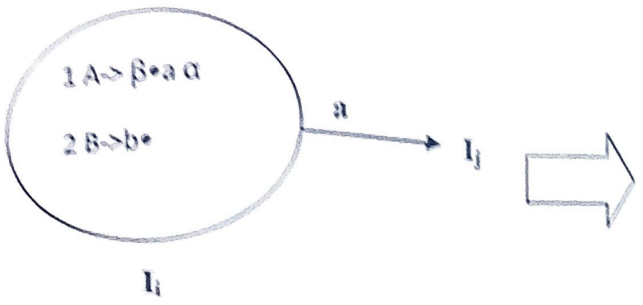
If there is a transaction from one state (I_i) to another state (I_j) on a Non terminal value then, we should write the subscript value of I_i in the GO TO part as shown below: part as shown below:



States	ACTION		GO TO
	a	$	A
I_i			j
I_j			

Shift-Reduce Conflict in SLR (1) Parsing : Shift Reduce Conflict in the LR (1) parsing occurs when a state has

1. A Reduced item of the form A → α• and Follow(A) includes the terminal value 'a'.
2. An incomplete item of the form A → β•aα as shown below:



States	Action		GOTO	
	a	$	A	B
I_i	Sj/r2			

Reduce - Reduce Conflict in SLR (1) Parsing

Reduce- Reduce Conflict in the LR (1) parsing occurs when a state has two or more reduced items of the form

1. **A → α•**
2. **B → β• and Follow (A) ∩ Follow(B) ≠ null** as shown below:

If The Grammar is

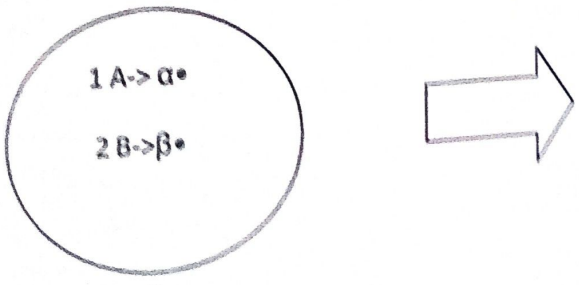
S-> αAaBa

A-> α

B-> β

Follow(S)= {$}

Follow(A)={a} and Follow(B)= {a}



States	Action		GOTO	
	a	$	A	B
I_i	$r_1/r2$			

Canonical LR (1) Parsing: Various steps involved in the CLR (1) Parsing:

1. Write the Context free Grammar for the given input string
2. Check for the Ambiguity
3. Add Augment production

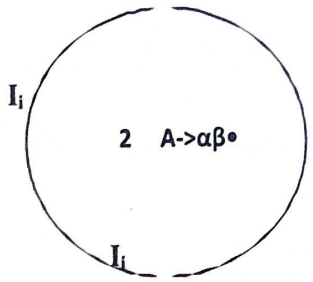
I_i I_j

1 If there is one state (I_i), where there is one production **(A->αβ•)** which has no transition to the next State. Then, the production is said to be a reduced production. For all terminals X in FOLLOW (A), write the reduce entry along with their production numbers. If the Augment production is reducing then write accept.

1 S -> •aAb

2 A->αβ•

Follow(S) = {$}
Follow (A) = (b}



States	ACTION			GO TO	
	a	b	$	S	A
I_i		r_2			

SLR (1) table for the Grammar

S → aB
B → bB | b

Follow (S) = {$}, Follow (B) = {$}

States	ACTION			GOTO	
	A	b	$	S	B
I_0	S_2			1	
I_1			ACCEPT		
I_2		S_4			3
I_3			R_1		
I_4		S_4	R_3		5
I_5			R_2		

Note: When Multiple Entries occurs in the SLR table. Then, the grammar is not accepted by SLR(1) Parser.

Conflicts in the SLR (1) Parsing :

When multiple entries occur in the table. Then, the situation is said to be a Conflict.

4. Create Canonical collection of LR (1) items
5. Draw DFA
6. Construct the CLR (1) Parsing table
7. Based on the information from the Table, with help of Stack and Parsing algorithm generate the output.

LR (1) items :

The LR (1) item is defined by **production, position of data** and a **terminal symbol**. The terminal is called as ***Look ahead symbol.***

General form of LR (1) item is

S->α•Aβ , $
A-> •γ, FIRST(β,$)

Rules to create canonical collection:

1. Every element of I is added to closure of I
2. If an LR (1) item **[X-> A•BC, a]** exists in I, and there exists a production B->b_1b_2...., then add item **[B->• b_1b_2, z]** where z is a terminal in **FIRST(Ca),** if it is not already in Closure(I).keep applying this rule until there are no more elements adde.

For example, if the grammar is

S->CC
C->cC
C->d

The Canonical collection of LR (1) items can be created as follows:

0. S'->•S (Augment Production)
1. S->•CC
2. C->•cC
3. C->•d

I_0 State : Add Augment production and compute the Closure, the look ahead symbol for the Augment Production is $.

S'->•S, $= Closure(S'->•S, $)

The dot symbol is followed by a Non terminal S. So, add productions starting with S in I_0 State.

S->•CC, FIRST ($), using 2nd rule

S->•CC, $

The dot symbol is followed by a Non terminal C. So, add productions starting with C in I State.

C->•cC, FIRST(C, $)
C->•d, FIRST(C, $)

FIRST(C) = {c, d} so, the items are

C->•cC, c/d
C->•d, c/d

The dot symbol is followed by a terminal value. So, close the I_0 State. So, the productions in I_0 are

S'->•S , $
S->•CC , $
C->•cC, c/d
C->•d , c/d

I_1 = Goto (I_0, S)= S'->S•,$

I_2 = Go to (I_0 , C)= Closure(S-> C•C, $)

S-> C->•cC , $
C->•d,$ So, the I_2 State is

S->C•C,$
C->•cC , $
C->•d,$

I_3= Goto(I_0,c)= Closure(C->c•C, c/d)
C->•cC, c/d
C->•d , c/d So, the I_3 State is

C->c•C, c/d
C->•cC, c/d
C->•d , c/d

I_4= Goto(I_0, d)= Colsure(C->d•, c/d) = C->d•, c/d

I_5 = Goto (I_2, C)= closure(S->CC•,$)= S->CC•, $

I_6= Goto (I_2, c)= closure(C->c•C , $)=
C->•cC, $
C->•d , $ SO, the I_6 State is